Auto Exam Development Status

Two Phase Completion Report

Gautam Buddha University
Assessment date: 2 October 2026
Basis: current local application code, schema, documentation, and executed tests.

Current position

The academic and examination-scheduling core is implemented. The available scheduling tests pass, including generation, review, locking, regeneration, approval, publication, and Excel export. Seating, attendance, invigilation, replacements, and reports already contain working features, but need additional controls, corrections, or acceptance testing before the complete examination platform can be called finished.

Assessment	Result
Core-complete module groups	6 of 14 reviewed groups
Partially complete module groups	7 of 14 reviewed groups
Remaining delivery workstream	1 of 14 groups: archival, recovery validation, and release hardening
Executed automated test scripts	17 total: 16 passed; 1 failed during setup
Scheduling conclusion	Complete for the implemented core workflow; broader roadmap features and production acceptance remain

How to read completion

Core complete means the stated core workflow exists in the inspected application. It does not mean every original roadmap enhancement is delivered or every browser workflow has been accepted. **Partial** means useful features exist but identifiable work remains. **Remaining** means the end-to-end delivery workstream has not been demonstrated, although supporting files or schema may exist.

The module count is not an effort-weighted project percentage: a small import feature and a large operations workflow do not require equal work. Likewise, 16 passing tests out of 17 does not mean the application is 94% complete.

Two proposed delivery phases

Phase 1 - Academic foundation and scheduling: recognise the completed core and demonstrate the scheduling workflow.
Phase 2 - Examination operations and release readiness: complete operational controls, resolve defects, and validate the whole examination cycle.

Phase 1 Academic Foundation and Scheduling

These six module groups form the completed core baseline. Completion here is scoped to the capabilities below; it is not a claim that all features in the earlier development plan are finished.

Module group	Completed core capabilities	Evidence
1. Academic masters and curriculum	School/programme records, admission batches, programme duration, batch/semester curriculum, course status, priority, and curriculum filters.	Seed, curriculum import, batch alias, batch progression, and course-deletion tests pass.
2. Student management	Record creation/editing, identifier parsing, student register, programme/batch filtering, search, and staged student imports.	StudentRepository, StudentImportService, student views; parser and eligibility scenarios pass. Full student CRUD/import browser acceptance not run.
3. Faculty master and availability	Faculty records, faculty import, active status, and date/shift availability.	FacultyImportService, faculty forms, availability routes. Duty operations assessed separately in Phase 2.
4. Examination cycles and calendar	Cycles, selected batches, shift definitions, examination dates, holidays, Sundays, and disabled dates.	ExamService and calendar views; scheduling tests pass. Cycle deletion is separately unresolved.
5. Scheduling and date sheets	School/batch/semester scope, paper selection, manual/import routes, automatic draft generation, conflicts, locks, moves, regeneration, comparison, approval/publication.	AutomaticScheduling, PreferredGapScheduling, and SoictScheduling tests pass.
6. Import validation foundation	CSV/XLSX reading, staging, row validation, previews, accepted-row commits, and error messages in the implemented import modules.	Programme, course, room, and curriculum dataset import tests pass; other import types need acceptance coverage.

Scheduling is complete within the current scope

The tested core supports priority ordering, exclusions for lab/practical papers, calendar restrictions, common-course scenarios, paper-duration checks, preserved locks, safe regeneration, approval, and publication. Batch progression and date-sheet Excel export also pass their available tests.

Scheduling is not a guarantee of every older roadmap item. Weekly limits, mandatory room-capacity checking during scheduling, advanced backtracking/optimisation, and a separate holiday bulk-upload workflow were not established by this review. They should be treated as extensions or confirmed scope decisions.

Important policy detail: the current scheduler can relax both the preferred rest gap and preferred daily paper count when the calendar is tight, using additional shifts. Same-cohort date/shift collisions remain prohibited. Staff must agree whether those preferences are acceptable before production use.

Phase 2 Examination Operations Status

Phase 2 completes and validates the operational chain after scheduling. These modules are not empty; most already have a usable starting implementation.

Module group	Status	Already developed	Still required
7. Authentication and access	Partial	Login/logout, password hashes, CSRF, sessions, eight roles, permission routing.	Reconcile security documentation with write permissions; verify protected browser actions for every role.
8. Rooms and physical layouts	Partial	Room import, row/column grids, blocked seats, usable capacity, priority, row/column order.	Advanced custom seat ordering/aisles if retained in scope; protection and regression checks for geometry edits.
9. Seating and eligibility	Partial	Eligibility-driven interleaving, versions, capacity checks, unallocated list, validation/publication.	Repair manual move query; enforce correction reason on server; validate published-version transitions and correction routes.
10. Attendance	Partial	Records from published seating, marking, six statuses, room sheets, totals and CSV.	Separate correction history and reasons, corrected-user/time fields, audit events, controlled finalisation.
11. Invigilation	Partial	One faculty per used room, unavailable/busy exclusions, duty-count ordering, duty list and CSV.	Occupancy-based staffing, reassignment controls, role-specific duty workflow, workload reconciliation and tests.
12. Replacements	Partial	Request form, nominated faculty, pending/approved/rejected states, reviewer/time.	Check replacement availability and valid candidate/reason; preserve original duty history; update workloads and audit changes.
13. Reports and audit	Partial	Dashboard, selected CSV exports, date-sheet XLSX, print views, audit page and several audited changes.	Attendance/duty/replacement audit coverage; final operational report packs and print validation; added formats only where agreed.
14. Archival recovery and release	To finish	Lifecycle status fields, deletion guards, backup/deployment guidance, tests and hosting scaffolding.	Prove closure/archive workflow, retention, restore drill, clean deployment, security/load/accessibility checks and UAT.

The basic seating and attendance scenarios in EndToEndTest pass, but that script directly writes publication and attendance records. It does not prove that the browser publication, attendance-edit, or correction routes work correctly.

Specific Gaps to Complete in Phase 2

1 Seating correction defect

The manual move handler selects **a.room_id** from the seating_allocations alias, but that table has no room_id column. The room reference belongs to the assignment. This is a code-level defect in the correction route, not a failure of the seating generation algorithm. The handler also supplies a default reason when no reason is posted, even though the form requires one. Correct the query and require an explicit server-side reason.

2 Attendance correction accountability

saveAttendance updates marked_by and marked_at when saving statuses. The schema includes corrected_by and corrected_at, but the inspected handler does not use them or preserve a separate correction event. Initial attendance marking works; a fully traceable correction workflow remains to be completed.

3 Invigilation staffing and workload

Allocation chooses one available faculty member for each occupied room and orders candidates by duty_count. It does not implement a configurable invigilator requirement based on room occupancy. Duty counts increment on allocation; replacement approval does not reconcile original/replacement workload counts. Staffing policy, reassignment, and workload lifecycle handling need completion.

4 Replacement approval controls

The approval path changes the faculty ID on the existing duty. It does not explicitly recheck unavailable status, validate that a replacement exists, or retain the original faculty ID on the replacement request. Database uniqueness provides a same-session conflict backstop, but it is not a substitute for complete candidate validation or historical traceability. Preserve original and new faculty details and record the approved change.

5 Access policy and audit consistency

SECURITY.md describes rejecting all authenticated non-administrator POST requests. RolePolicy and inspected routing instead permit operations by module permissions. Align the agreed access policy, implementation, documentation, and browser tests. Attendance, duty allocation, and replacement handlers also need consistent audit events.

6 Cycle deletion regression and release proof

ExamCycleDeletionTest stops while preparing an examination with the message: "The selected subject is not assigned to this batch and semester curriculum." The fixture selects a mapping without passing its batch context to scheduling. This suggests a fixture/context mismatch; it does not establish that the deletion service itself is broken. Correct the setup and rerun the deletion assertions.

Closure/archive status fields and documentation alone do not prove an archival workflow. Complete or confirm the intended user actions, retained snapshots, retention policy, restoration drill, and production acceptance evidence.

Verification Results and Evidence

Tests were executed locally on 2 October 2026 with PHP assertions enabled: **zend.assertions=1** and **assert.exception=1**. Database-backed tests use their isolated temporary databases. No application code was changed as part of this status review.

Test script	Result	What the result supports
RollNumberParserTest	PASS	Six identifier parsing cases.
RolePolicyTest	PASS	Twelve permission checks across eight role definitions.
BatchProgressionTest	PASS	Odd/even semester and batch completion rules.
WrittenPaperPolicyTest	PASS	Lab/practical recognition and boundaries.
XlsxExporterTest	PASS	Valid programme/batch/semester date-sheet workbook.
AutomaticSchedulingTest	PASS	Selection, priority, holidays, gaps, locks, regeneration, approval/publication.
PreferredGapSchedulingTest	PASS	Spacing preferences and adaptive use of multiple shifts.
SoictSchedulingTest	PASS	Branch rows, shared papers, locks, publication, duration checks.
SeedTest	PASS	Repeatable seed, eight roles and curriculum counts.
CourseImportTest	PASS	Curriculum validation, staging and commit.
CourseBatchLabelAliasTest	PASS	Short batch-year alias resolution.
ProgrammeImportTest	PASS	Programme import validation and commit.
RoomImportTest	PASS	Room import, geometry and disabled seats.
BTechCseCurriculumDatasetTest	PASS	Dataset commit and blank-row rejection.
EndToEndTest	PASS	Seating, shared papers, eligibility, capacity and direct attendance creation.
CourseDeletionTest	PASS	Dependency protection and course deletion audit checks.
ExamCycleDeletionTest	FAIL	Curriculum-context exception during setup; deletion assertions not reached.

Passing tests demonstrate their specific scenarios. They do not establish complete browser acceptance, production hosting reliability, full security coverage, all possible scheduling datasets, accessibility conformance, or measured load capacity. Student/faculty import acceptance and duty/replacement workflows need stronger dedicated coverage.

Phase 2 Delivery Plan and Completion Gates

Step	Work package	Completion condition
1	Resolve known defects and policy differences	Manual seating moves work; reasons are required on the server; cycle-deletion test reaches its assertions and passes; access policy is consistent.
2	Finish attendance controls	Initial marking and corrections retain actor/time, before/after status and reason; published-session access is enforced; correction tests pass.
3	Finish invigilation and replacements	Agreed staffing rules are enforced; unavailable/double-booked candidates are rejected; original duty history and workloads remain correct after replacement.
4	Complete reports and audit	The examination cell can produce the agreed report pack; exports reconcile with records; attendance, duty and replacement changes are auditable.
5	Complete lifecycle and recovery	Closure/archive actions and retention policy are demonstrated; a backup is restored successfully; operational history remains available.
6	Accept the full release	A representative cycle passes browser UAT from imports to final attendance/duties, alongside agreed load, accessibility, security and clean-hosting checks.

Recommended project statement

"Phase 1 is complete for the academic foundation and core examination scheduling workflow. Phase 2 is partially implemented and covers completion of seating corrections, attendance accountability, invigilation, replacement controls, reporting, archival, and production validation."

This statement gives scheduling proper credit without incorrectly claiming that other modules have not been developed. Production sign-off should follow the Phase 2 gates rather than a guessed percentage.

Source files reviewed

README.md; DEVELOPMENT_PLAN.md; plan.md; SECURITY.md; database/schema.sql; app/Auth/RolePolicy.php; app/Foundation/Application.php; app/Academic/StudentRepository.php; app/Rooms/RoomSeatGenerator.php; app/Seating/SeatingAllocator.php; app/Exams/AutomaticScheduler.php; SchedulingValidator.php; SchedulePublicationService.php; relevant module views and the 17 test scripts listed on the previous page.

Assessment boundaries

This is a code-and-test status assessment of the current local checkout. It is not a live browser demonstration or a formal SRS acceptance audit. Earlier planning documents contain features and rule definitions that differ from the implemented scheduler; the current code and executed tests take precedence for this report. Advanced roadmap enhancements should be explicitly retained, revised, or removed from the agreed scope.